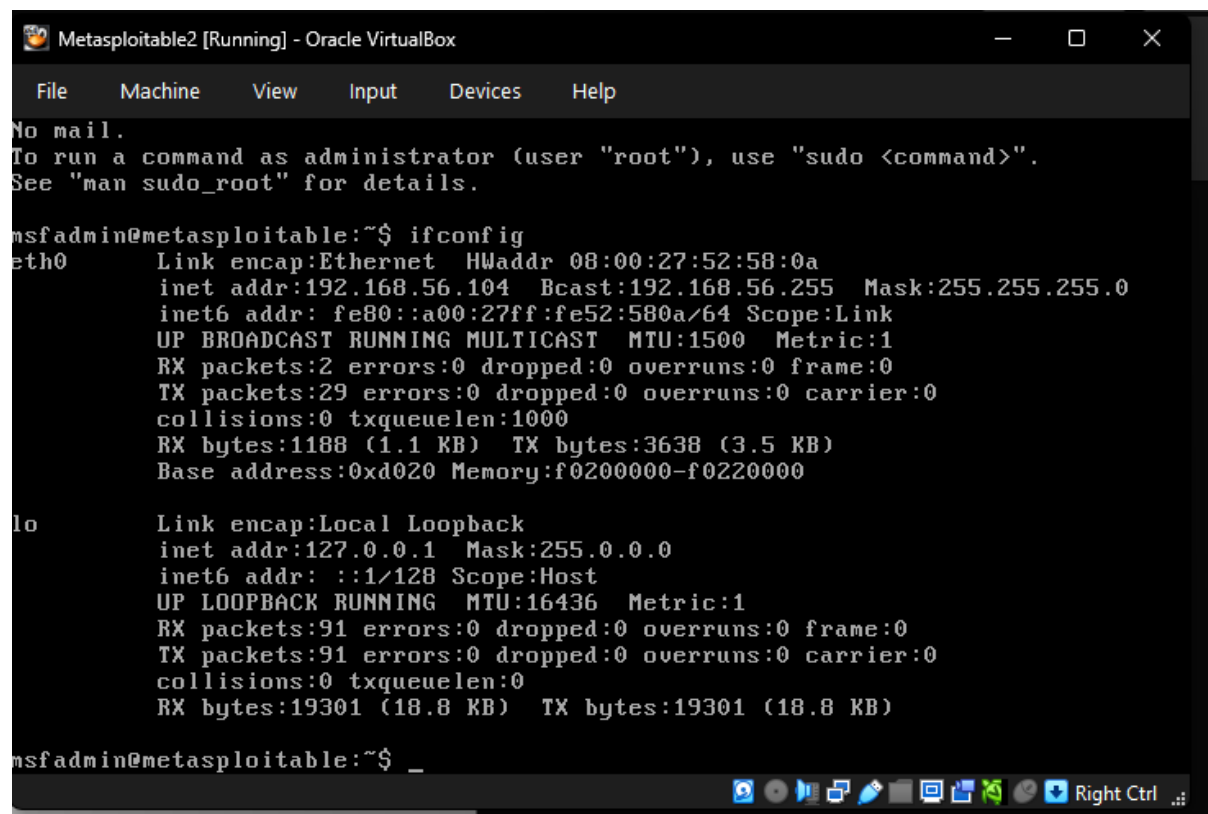


Summary Table

No.	Challenge	Tool / Command	Key Finding
C1	NetBIOS Enumeration	nmblookup -A	Hostname METASPLOITABLE, SMB aktif (<20>), Workgroup: WORKGROUP
C2	Fast Nmap Scan	nmap -F	10+ port terbuka: FTP (21), Telnet (23), MySQL (3306), SMB (445)
C3	DNS Records	dig / nslookup	A, AAAA, MX, NS records berjaya diperolehi dari domain awam
C5	TTL OS Fingerprinting	ping	TTL=64 — confirmed Linux/Unix target (Metasploitable 2)
C7	SMTP VRFY / EXPN	nc port 25	VRFY root & admin → 252 (user exists); Server: Postfix Ubuntu
C9	FTP Banner	nc port 21	vsFTPD 2.3.4 — Backdoor CVE-2011-2523 (CVSS 10.0 CRITICAL)
C10	Anonymous FTP Login	ftp	Login anonymous berjaya — direktori /pub boleh diakses tanpa auth
C11	SMB NSE Enumeration	nmap --script	Samba 3.0.20 (CVE-2007-2447 RCE), 4 user accounts didedahkan
C16	Version Detection	nmap -sV	Semua service versi lapuk: Apache 2.2.8, MySQL 5.0.51a, OpenSSH 4.7p1
C17	OS Detection	nmap -O	OS: Linux kernel 2.6.x, 20+ port terbuka dikesan

Installing Metasploitable 2



```
Metasploitable2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

No mail.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:52:58:0a
          inet addr:192.168.56.104  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe52:580a/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1188 (1.1 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$ _
```

Challenge 1 — NetBIOS Enumeration

nmblookup -A 192.168.56.104

```
(kali㉿kali)-[~]  
$ nmblookup -A 192.168.56.104  
Looking up status of 192.168.56.104  
    METASPLOITABLE <00> - B <ACTIVE>  
    METASPLOITABLE <03> - B <ACTIVE>  
    METASPLOITABLE <20> - B <ACTIVE>  
    WORKGROUP <00> - <GROUP> B <ACTIVE>  
    WORKGROUP <1e> - <GROUP> B <ACTIVE>  
  
    MAC Address = 00-00-00-00-00-00
```

Challenge 2 — Fast Nmap Scan

nmap -F 192.168.56.104

```
(kali㉿kali)-[~]  
$ nmap -F 192.168.56.104  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-03 02:48 -0400  
Nmap scan report for 192.168.56.104  
Host is up (0.0019s latency).  
Not shown: 82 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
513/tcp   open  login  
514/tcp   open  shell  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
8009/tcp  open  ajp13  
  
Nmap done: 1 IP address (1 host up) scanned in 2.23 seconds
```

Challenge 3 — DNS Records

nslookup google.com

```
(kali@kali)-[~]  
$ nslookup google.com  
Server:      192.168.100.1  
Address:     192.168.100.1#53  
  
Non-authoritative answer:  
Name:   google.com  
Address: 142.250.193.238  
Name:   google.com  
Address: 2404:6800:4001:804::200e
```

dig ANY google.com

```
(kali@kali)-[~]  
$ dig ANY google.com  
  
; <<>> DiG 9.20.20-1-Debian <<>> ANY google.com  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 16921  
;; flags: qr rd ra; QUERY: 1, ANSWER: 8, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
;; QUESTION SECTION:  
;google.com.                IN      ANY  
  
;; ANSWER SECTION:  
google.com.                72      IN      AAAA    2404:6800:4001:804::200e  
google.com.                20490   IN      HTTPS   1 . alpn="h2,h3"  
google.com.                27      IN      SOA     ns1.google.com. dns-admin.google.com. 909143293 900 900 1800 60  
google.com.                51      IN      A       142.250.193.238  
google.com.                34945   IN      NS      ns4.google.com.  
google.com.                34945   IN      NS      ns3.google.com.  
google.com.                34945   IN      NS      ns1.google.com.  
google.com.                34945   IN      NS      ns2.google.com.  
  
;; Query time: 11 msec  
;; SERVER: 192.168.100.1#53(192.168.100.1) (TCP)  
;; WHEN: Sun May 03 02:51:11 EDT 2026  
;; MSG SIZE rcvd: 226
```

dig MX google.com

```
(kali㉿kali)-[~]  
$ dig MX google.com  
  
; <<>> DiG 9.20.20-1-Debian <<>> MX google.com  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 11527  
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags;; udp: 1232  
;; QUESTION SECTION:  
;google.com.                IN      MX  
  
;; ANSWER SECTION:  
google.com.                300     IN      MX      10 smtp.google.com.  
  
;; Query time: 19 msec  
;; SERVER: 192.168.100.1#53(192.168.100.1) (UDP)  
;; WHEN: Sun May 03 02:51:21 EDT 2026  
;; MSG SIZE  rcvd: 60
```

Challenge 5 — TTL OS Fingerprinting

Ping 192.168.56.104

```
(kali㉿kali)-[~]  
$ ping 192.168.56.104  
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data.  
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=1.74 ms  
64 bytes from 192.168.56.104: icmp_seq=2 ttl=64 time=1.42 ms  
64 bytes from 192.168.56.104: icmp_seq=3 ttl=64 time=1.23 ms  
64 bytes from 192.168.56.104: icmp_seq=4 ttl=64 time=1.33 ms  
64 bytes from 192.168.56.104: icmp_seq=5 ttl=64 time=1.27 ms  
64 bytes from 192.168.56.104: icmp_seq=6 ttl=64 time=1.83 ms  
64 bytes from 192.168.56.104: icmp_seq=7 ttl=64 time=1.18 ms  
64 bytes from 192.168.56.104: icmp_seq=8 ttl=64 time=1.18 ms  
^C
```

Challenge 7 — SMTP VRFY / EXPN

echo -e "VRFY root\nVRFY admin\nVRFY nobody\nEXPN root\nQUIT" | nc 192.168.56.104 25

```
(kali㉿kali)-[~]  
$ echo -e "VRFY root\nVRFY admin\nVRFY nobody\nEXPN root\nQUIT" | nc 192.168.56.104 25  
220 metasploitable.localdomain ESMTP Postfix (Ubuntu)  
252 2.0.0 root  
550 5.1.1 <admin>: Recipient address rejected: User unknown in local recipient table  
252 2.0.0 nobody  
502 5.5.2 Error: command not recognized  
221 2.0.0 Bye
```

Challenge 9 — FTP Banner

nc 192.168.56.104 21

```
(kali㉿kali)-[~]  
$ nc 192.168.56.104 21  
220 (vsFTPd 2.3.4)  
quit  
221 Goodbye.
```

Challenge 10 — Anonymous FTP Login

ftp 192.168.56.104

```
(kali@kali)-[~]  
$ ftp 192.168.56.104  
Connected to 192.168.56.104.  
220 (vsFTPd 2.3.4)  
Name (192.168.56.104:kali): anonymous  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> ls  
229 Entering Extended Passive Mode (|||55470|).  
150 Here comes the directory listing.  
226 Directory send OK.  
ftp> pwd  
Remote directory: /  
ftp> bye  
221 Goodbye.
```

Challenge 11 — SMB NSE Enumeration

nmap --script smb-os-discovery -p445 192.168.56.104

```
(kali@kali)-[~]  
$ nmap --script smb-os-discovery -p445 192.168.56.104  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-03 03:14 -0400  
Nmap scan report for 192.168.56.104  
Host is up (0.00047s latency).  
  
PORT      STATE SERVICE  
445/tcp   open  microsoft-ds  
  
Host script results:  
| smb-os-discovery:  
|   OS: Unix (Samba 3.0.20-Debian)  
|   Computer name: metasploitable  
|   NetBIOS computer name:  
|   Domain name: localdomain  
|   FQDN: metasploitable.localdomain  
|_  System time: 2026-05-03T03:14:00-04:00  
  
Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds
```

nmap --script smb-enum-users -p445 192.168.56.104

```
(kali㉿kali)-[~]
└─$ nmap --script smb-enum-users -p445 192.168.56.104
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-03 03:14 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00056s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds

Host script results:
| smb-enum-users:
|   METASPLOITABLE\backup (RID: 1068)
|     Full name:  backup
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\bin (RID: 1004)
|     Full name:  bin
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\bind (RID: 1210)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\daemon (RID: 1002)
|     Full name:  daemon
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\dhcp (RID: 1202)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\distccd (RID: 1222)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\ftp (RID: 1214)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\games (RID: 1010)
|     Full name:  games
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\gnats (RID: 1082)
|     Full name:  Gnats Bug-Reporting System (admin)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\irc (RID: 1078)
|     Full name:  ircd
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\klog (RID: 1206)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\libuuid (RID: 1200)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\list (RID: 1076)
|     Full name:  Mailing List Manager
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\lp (RID: 1014)
|     Full name:  lp
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\mail (RID: 1016)
|     Full name:  mail
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\man (RID: 1012)
```


Challenge 16 — Version Detection

nmap -sV 192.168.56.104

```
(kali@kali)~$ nmap -sV 192.168.56.104
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-03 03:16 -0400
Nmap scan report for 192.168.56.104
Host is up (0.0027s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login?         Netkit rshd
514/tcp   open  shell          GNU Classpath grmiregistry
1099/tcp  open  java-rmi        GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.
c917new-service :
SF-Port513-TCP:-V-7.99XI-7KD-5/3%Time-69F6F655XP-x86_64-pc-linux-gnuKr(DNSV
SF:ersionBindReqTCP,1,"x01")Kr(DNSStatusRequestTCP,1,"x01")Kr(Kerberos,1
SF:,"x01")Kr(SMBProgNeg,1,"x01")Kr(oracle-tns,1,"x01")Kr(afp,1,"x01");
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 162.58 seconds
```

Challenge 17 — OS Detection

nmap -O 192.168.56.104

```
(kali@kali)~$ nmap -O 192.168.56.104
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-03 02:57 -0400
Nmap scan report for 192.168.56.104
Host is up (0.00096s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BCM210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.24 seconds
```